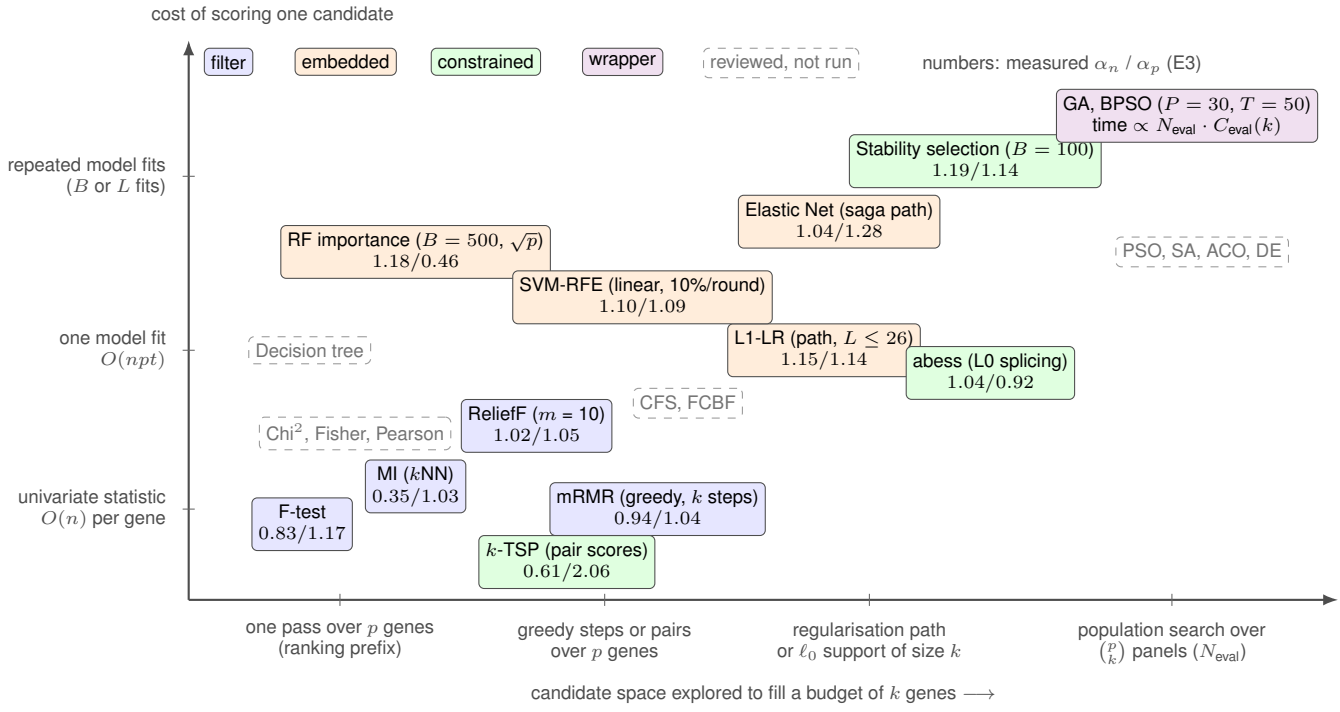

cost of scoring one candidate
filter
embedded
constrained
wrapper
reviewed, not run
numbers: measured α_n / α_p (E3)
GA, BPSO ($P = 30$, $T = 50$)
time $\propto N_{\text{eval}} \cdot C_{\text{eval}}(k)$
Stability selection ($B = 100$)
1.19/1.14
Elastic Net (saga path)
1.04/1.28
RF importance ($B = 500$, $\sqrt{p}$)
1.18/0.46
PSO, SA, ACO, DE
SVM-RFE (linear, 10%/round)
1.10/1.09
repeated model fits
(B or L fits)
L1-LR (path, $L \leq 26$)
1.15/1.14
abess (L0 splicing)
1.04/0.92
one model fit
$O(npt)$
Decision tree
CFS, FCBF
ReliefF (m = 10)
1.02/1.05
Chi^2, Fisher, Pearson
MI (kNN)
0.35/1.03
univariate statistic
$O(n)$ per gene
mRMR (greedy, k steps)
0.94/1.04
F-test
0.83/1.17
k-TSP (pair scores)
0.61/2.06
one pass over p genes
(ranking prefix)
greedy steps or pairs
over p genes
regularisation path
or ℓ_0 support of size k
population search over
$\binom{p}{k}$ panels (N_{eval})
candidate space explored to fill a budget of k genes ⟶